

Speak Like A Mathematician:

What changes from congruence (Ch. 5) to similarity?

Why does AA need only two pairs of angles?

Three Ways to Prove Triangles Similar

AA Similarity: two pairs of congruent _____ make the triangles similar.

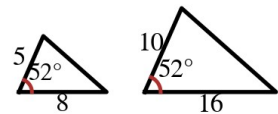
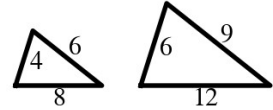
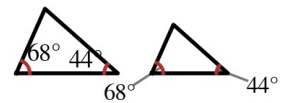
SSS Similarity: all three pairs of corresponding sides are _____.

SAS Similarity: two pairs of proportional sides and a congruent _____ angle.

Example 1 - Similar? Say Why (part 1)

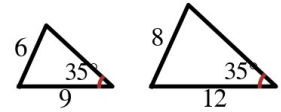
For each pair at the right, decide whether the triangles are similar. If they are, name the postulate or theorem.

- First pair (two marked angles each).
- Second pair (sides 4, 6, 8 and 6, 9, 12).
- Third pair (sides 5, 8 with a 52° included angle; sides 10, 16 with a 52° included angle).



Example 2 - Similar? Say Why (part 2)

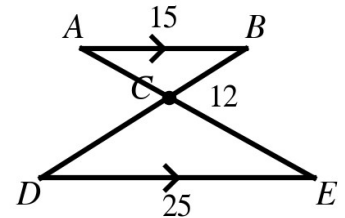
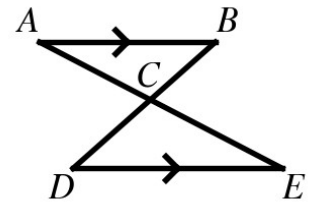
- a) Fourth pair (angles 75° , 40° and 75° , 38°).
- b) Fifth pair (sides 9, 6 and 12, 8 with a congruent angle NOT between them).
- c) Sixth pair ($\overline{AB} \parallel \overline{DE}$ crossing at C).



Example 3 - Overlapping Triangles

In the figure at the right, $\overline{AB} \parallel \overline{DE}$, $AB = 15$, $DE = 25$, and $BC = 12$.

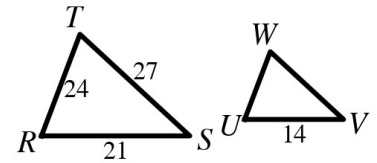
- a) Explain why $\triangle ABC \sim \triangle EDC$.
- b) Find the scale factor of $\triangle ABC$ to $\triangle EDC$ and the length CD.



Example 4 - Use Similarity to Find Sides

In the figure at the right, $\triangle RST \sim \triangle UVW$, $RS = 21$, $ST = 27$, $TR = 24$, and $UV = 14$.

- Find the scale factor of $\triangle RST$ to $\triangle UVW$.
- Find VW and WU .
- Find both perimeters. What is their ratio?



Example 5 - Indirect Measurement

A 6-ft-tall coach casts a 9-ft shadow. At the same time, the goalpost casts a 45-ft shadow.

- Why are the two triangles similar?
- Find the height of the goalpost.
- Why is this called INDIRECT measurement?

